

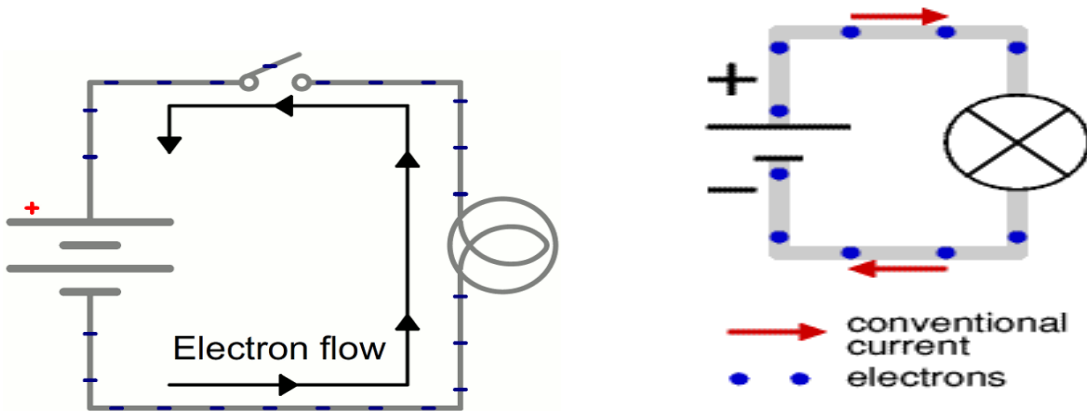
Chapter 3: DC Circuits and Electromagnetism

1.1 Introduction:

An electrical system generally transmits energy due to the movement of electric charge. Electricity appears in one of two forms which, by convention are called negative and positive electricity. Electric charge is the excess of negative or positive electricity on a body or in space. If the excess is negative, the body is said to have a negative charge and vice versa. An electron is an elementary particle charged with a small and constant quantity of negative electricity. A proton is similarly defined but charged with positive electricity whereas the neutron is uncharged and is therefore neutral.

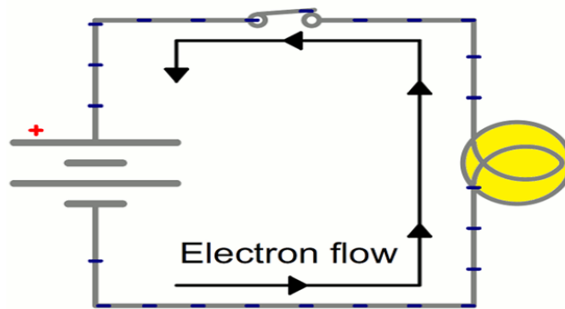
All electrons have a certain potential energy. Given a suitable medium to exist, they move freely from one energy level to another and this movement, when undertaken in a concerted manner, is termed as an electric current flow. Conventionally it is said that the current flows from a point of high energy level to a point of low energy level. These points are said to have high potential and low potential respectively. For convenience the point of high potential is termed the positive and the point of low potential is termed the negative, hence conventionally a current is said to flow from positive to negative. Unfortunately it was found that electrons move in the other direction since the negatively charged electron is attracted to the positive potential. Thus conventional current flows in the opposite direction to that of electron current.

The transfer of electrons takes place more readily in a medium in which atoms can readily release electrons; such a material is termed a conductor e.g. copper, aluminum, silver. A material that does not readily permit electron flow is termed an insulator, e.g. porcelain, nylon, rubber.



The following conditions are to be fulfilled for the flow of current:

1. There must be a complete circuit around which the electrons may move. If the electrons cannot return to the point of starting, then eventually they will all congregate together and the flow will cease.
2. There must be a driving influence to cause the continuous flow. This influence is provided by the source which causes the current to leave at a high potential and to move round the circuit until it returns to the source at a low potential.



The driving influence is termed the electromotive force, called the e.m.f. The e.m.f. represents the driving influence that causes a current to flow. The e.m.f. is not a force, but represents the energy expended during the passing of a unit charge through the source; an e.m.f. is always connected with energy conversion. The energy introduced into a circuit is transferred to the load unit by the transmission system, and the energy transferred due to the passage of unit charge between two points in a circuit is termed the potential difference (p.d.).

1.2 Important Terminologies:

1.2.1. Charge: Electric charge is the physical property of matter that causes it to experience a force when placed in an electromagnetic field

Symbol: Q

Unit: Coulomb (C)

1.2.2. Current: An electric current is a flow of electric charge or electron. The strength of the electric current is the rate of flow of electrons (i.e.,) charge flowing per second.

$$\text{Electric Current (I)} = \frac{dQ}{dt} = \frac{Q}{t}$$

Where, charge is in coulomb, time is in seconds, the current is in Ampere.

One ampere of current is said to flow through a wire if, at any section, one coulomb of charge flows in a second.

Symbol: I

Unit: Ampere (A)

1.2.3. Electric Potential: Electrical potential is defined as the capability of the charged body to do work. When the body is charged, either electrons are supplied to it, or they are removed from it. In both the cases, the work is done. This work is stored in the body in the form of electric potential.

$$\text{Electric Potential (V)} = \frac{\text{Work done}}{\text{Charge}} = \frac{W}{Q}$$

Where, Work done is in Joules, the charge is in coulombs, the electric potential is in Volts.

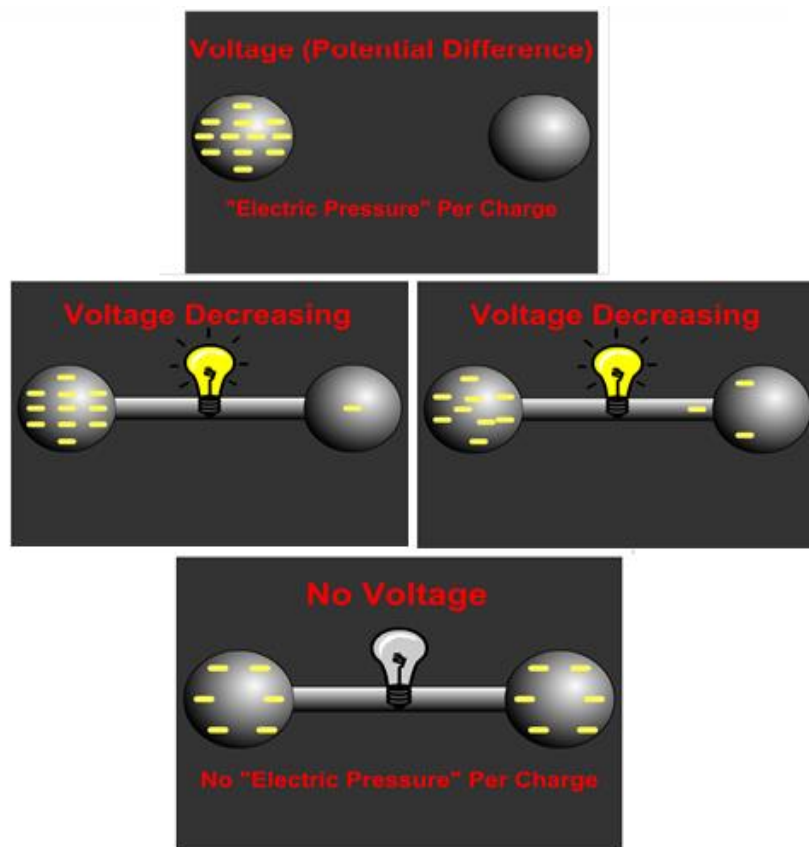
A body is said to have an electric potential of 1V if 1J of work is done to give it a charge of 1C

Symbol: V
Unit: Volt (V)

1.2.4. Potential difference: Potential difference is the difference in the amount of energy that charge carriers have between two points in a circuit. In other words, the potential difference is defined as the difference in the electric potential of the two charged bodies.

The potential difference between two points is 1 V if 1 J of work is done in transferring 1 C of charge from one point to another

Symbol: V
Unit: Volt (V)



1.2.5. Electromotive force: An electromotive force (e.m.f) is that which tends to produce an electric current in a circuit.

The potential difference causes current flow while an emf maintains the potential difference.

EMF is the energy supplied to the charge by an active device such as a battery

Symbol: E
Unit: Volt (V)

1.3 Fundamental linear passive and active elements

1.3.1 Linear circuit: A **linear circuit** is an electric circuit in which circuit parameters (Resistance, inductance, capacitance, waveform, frequency etc) are constant. In other words, a circuit whose parameters are not changed with respect to Current and Voltage is called Linear Circuit. Fundamentally, the word “linear” literally means “along with a straight line”. As the name says, a linear circuit means linear characteristics in between Current and Voltage, which means, current flowing through a circuit is directly proportional to the applied Voltage. If we increase the applied voltage, then the current flowing through the circuit will also increase, and vice versa. If we draw the circuit output characteristic curve in between Current and Voltage, it will look like a straight line (Diagonal). **Examples:** Resistors, Capacitors, Inductors, and Transformers etc.

1.3.2 Nonlinear circuit: A **nonlinear circuit** is an electric circuit whose parameters are varied with respect to Current and Voltage. In other words, an electric circuit in which circuit parameters (Resistance, inductance, capacitance, waveform, frequency etc) is not constant, is called Non Linear Circuit. If we draw the circuit output characteristic curve in between Current and Voltage, it will look like a curved or bending line. **Examples:** Diodes, Transistors, Iron core inductors etc.

1.3.3 Active components: Active components are the elements or devices which are capable of providing or delivering energy to the circuit hence they are also called as energy donor. They deliver or produce energy or power in the form of a voltage or current. They are capable of providing the power gain. Active components can control the flow of current. Active component requires an external source for the operation.

Examples: Diodes, Transistors, SCR, Integrated circuits (ICs) etc.

1.3.4 Passive components: Passive components are the devices which do not require any external source for the operation and are capable of storing energy in the form of voltage or current in the circuit. They utilize or store energy in the form of voltage or current hence they are also called as energy acceptor. They are not capable of providing the power gain. Passive components cannot control the flow of the current.

Examples: Resistor, Capacitor and Inductor.

1.4 Terminologies of an Electrical Network

1.4.1 Electrical network: An arrangement of electrical energy sources and circuit elements is called an electrical network. Electrical energy sources can be batteries and generators. Circuit elements are resistance (R), inductance (L), and capacitance (C).

1.4.2 Electrical Circuit: It is a closed path followed by an electric current. All electrical circuits are electrical networks.

1.4.3 Source: It is the device that supplies electrical energy. Example: Battery, Generator.

1.4.4 Load: It is the device that utilizes electrical energy. Example: Lamp, Motor, Heater.

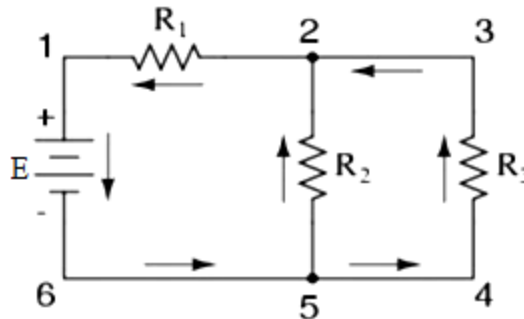


Figure A: Electrical Network or Electrical Circuit

1.4.5 Network elements: It is any circuit element with two terminals that can be connected to other circuit elements. Example: Resistors R_1 , R_2 , R_3 , source-E in Figure A.

1.4.6 Node: It is a junction where two or more circuit elements are connected together. Example: Points 1, 2, 3, 4, 5, 6 in Figure A.

1.4.7 Branch: It is the part of a circuit that lies between two junction points. Example: 1-2, 2-5, 1-6 in Figure A.

1.4.8 Loop: It is any closed path of a circuit. Example: 1-2-3-4-5-6-1, 1-2-5-6-1, 2-3-4-5-2 in Figure A.

1.4.9 Mesh: It is the most elementary form of a loop. A mesh cannot be further divided into other loops. Example: 1-2-5-6-1, 2-3-4-5-2 in Figure A. The loop 1-2-3-4-5-6-1 is not a mesh because it encloses two loops (1-2-5-6-1, 2-3-4-5-2).

1.5 Voltage Source and Current Source

Active Elements are the network elements that deliver power to other elements present in an electric circuit. So, active elements are also called as sources of voltage or current type. We can classify these sources into the following two categories –

- Independent Sources
- Dependent Sources

1.5.1 Independent Sources

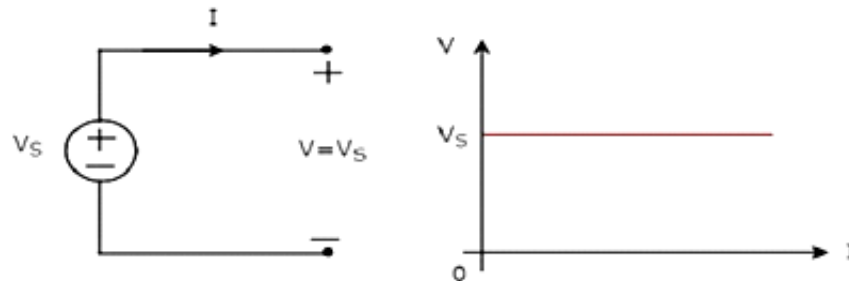
As the name suggests, independent sources produce fixed values of voltage or current and these are not dependent on any other parameter. Independent sources can be further divided into the following two categories –

- Independent Voltage Sources
- Independent Current Sources

1.5.1.1. Independent Voltage Sources

An independent voltage source produces a constant voltage across its two terminals. This voltage is independent of the amount of current that is flowing through the two terminals of voltage source.

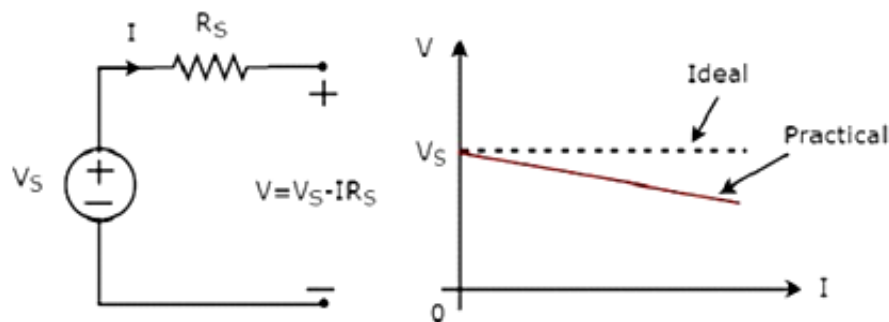
Independent **ideal voltage source** and its V-I characteristics are shown in the following figure.



The **V-I characteristics** of an independent ideal voltage source is a constant line, which is always equal to the source voltage (V_S) irrespective of the current value (I). So, the internal resistance of an independent ideal voltage source is zero Ohms.

Hence, the independent ideal voltage sources **do not exist practically**, because there will be some internal resistance.

Independent **practical voltage source** and its V-I characteristics are shown in the following figure.

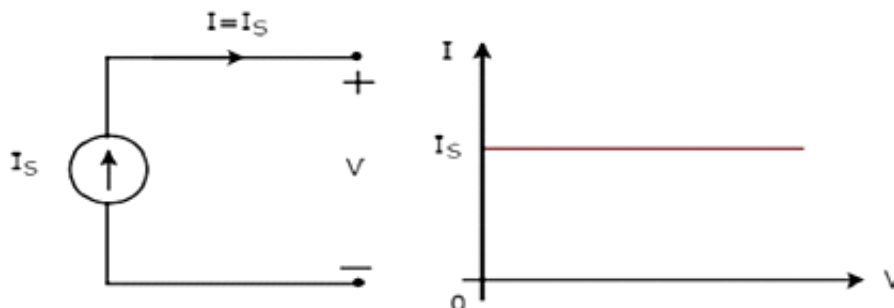


There is a deviation in the V-I characteristics of an independent practical voltage source from the V-I characteristics of an independent ideal voltage source. This is due to the voltage drop across the internal resistance (R_S) of an independent practical voltage source.

1.5.1.2. Independent Current Sources

An independent current source produces a constant current. This current is independent of the voltage across its two terminals.

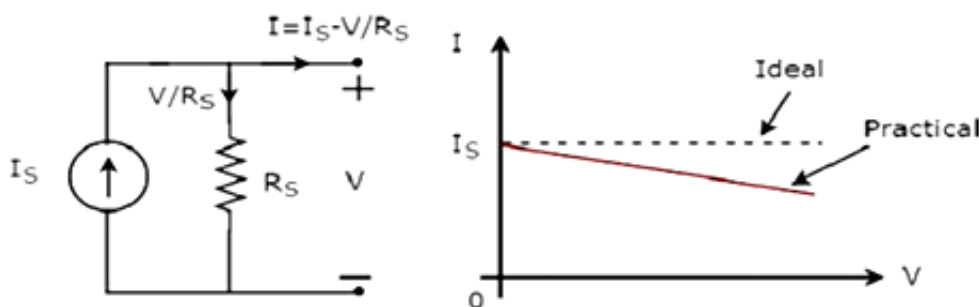
Independent **ideal current source** and its V-I characteristics are shown in the following figure.



The **V-I characteristics** of an independent ideal current source is a constant line, which is always equal to the source current (I_S) irrespective of the voltage value (V). So, the internal resistance of an independent ideal current source is infinite ohms.

Hence, the independent ideal current sources **do not exist practically**, because there will be some internal resistance.

Independent **practical current source** and its V-I characteristics are shown in the following figure.



There is a deviation in the V-I characteristics of an independent practical current source from the V-I characteristics of an independent ideal current source. This is due to the amount of current flows through the internal shunt resistance (R_S) of an independent practical current source.

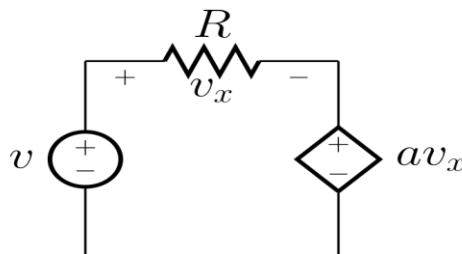
1.5.2 Dependent Voltage and Current Source: The sources whose output voltage or current is not fixed but depends on the voltage or current in another part of the circuit is called Dependent or Controlled source. They are four terminal devices. When the strength of voltage or current changes in the source for any change in the connected network, they are called dependent sources. The dependent sources are represented by a diamond shape.

The dependent sources are further categorized as :

1.5.2.1. Voltage Controlled Voltage Source (VCVS): The source delivers the voltage as per the voltage of the dependent element.

$$V = f_a(v_x)$$

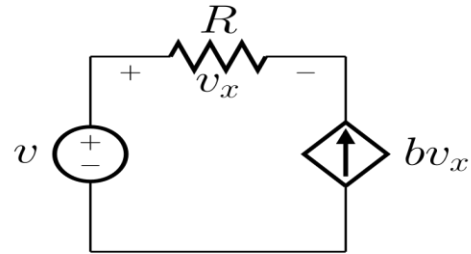
Where f_a is a constant



1.5.2.2. Voltage Controlled Current Source (VCCS): The source delivers the current as per the voltage of the dependent element

$$I = f_b(v_x)$$

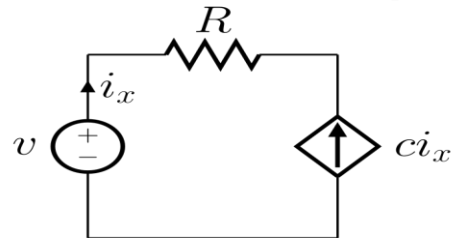
Where f_b is a constant



1.5.2.3. Current Controlled Current Source (CCCS): The source delivers the current as per the current of the dependent element.

$$I = f_c(i_x)$$

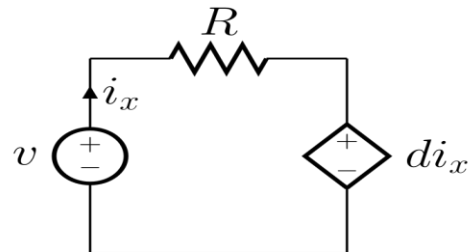
Where f_c is a constant



1.5.2.4. Current Controlled Voltage Source (CCVS): The source delivers the voltage as per the current of the dependent element.

$$V = f_d(i_x)$$

Where, f_d is a constant.



1.7 Ohm's Law:

Ohm's law states that the potential difference across the ends of a conductor is directly proportional to the current flowing between them provided that temperature and other physical quantity remain constant. Since the symbol for potential difference is V and current is I , this relationship may be expressed as

$$V \propto I$$

This is the expression of Ohm's law.

Since the relationship is assumed constant, we can write

$$V = I \cdot R$$

Where, R is a constant termed the resistance of the conductor.

It should be noted that this relationship is derived from Ohm's law and is not a symbolic expression for it. Ohm's law only notes the constancy of p.d. to current provided that other physical factors remain unchanged, i.e. for a given p.d. the current will vary in consequence of variation of external physical factors.

Limitations of Ohm's Law

1. The law is not applicable to unilateral networks.

Unilateral networks allow the current to flow in one direction. Such types of network consist of elements like a diode, transistor, etc.

2. Ohm's law is also not applicable to non – linear elements.

Non-linear elements are those which do not have current exactly proportional to the applied voltage that means the resistance value of those elements changes for different values of voltage and current. Examples of non – linear elements are the thyristor.

3. The relation between V and I depends on the sign of V.

In other words, if I is the current for a certain V, then reversing the direction of V keeping its magnitude fixed, does not produce a current of the same magnitude as I in the opposite direction. This happens for example in the case of a diode.

1.8 Kirchhoff's laws

We know that in a series circuit, the sum of the voltages across each of the components is equal to the applied voltage; and the sum of the currents in the branches of a parallel network is equal to the supply current. Gustav Kirchhoff, a German physicist, observed that these were particular instances of two general conditions fundamental to the analysis of any electrical network. These conditions may be stated as follows:

First (current) law: At any instant the algebraic sum of the currents at a junction in a network is zero. Different signs are allocated to currents flowing towards the junction and to those flowing away from it.

Second (voltage) law: At any instant in a closed loop, the algebraic sum of the e.m.f.s acting round the loop is equal to the algebraic sum of the potential differences round the loop.

1.8.1 Kirchhoff Current Law: It states that at any instant the algebraic sum of the currents at a junction in a network is zero.

Example: Consider a simple example as shown in figure,

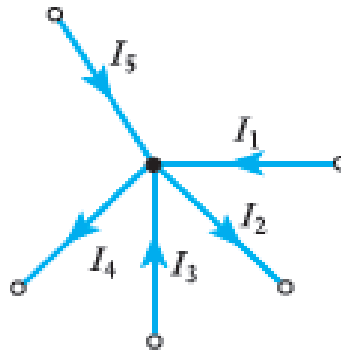


Illustration of Kirchhoff Current Law

Let the currents flowing towards the junction be considered positive and those flowing away from the junction are negative, hence according to the statement the equation can be written as:

$$I_1 - I_2 + I_3 - I_4 + I_5 = 0$$

OR

$$I_1 + I_3 + I_5 = I_2 + I_4$$

1.8.2 Kirchhoff Voltage Law: It states that at any instant in a closed loop, the algebraic sum of the emfs acting round the loop is equal to the algebraic sum of the potential differences round the loop.

(OR)

The algebraic sum of the products of currents and resistances in each of the conductors in any closed path (or mesh) in a network plus the algebraic sum of the emfs in that path is zero.

$$\sum IR + \sum \text{emf} = 0 \dots\dots\dots \text{round the mesh}$$

Assumptions/Sign conventions:

- i) Rise in voltage (If we go from negative terminal of the battery to positive terminal) = positive
- ii) Fall in voltage (If we go from positive terminal of the battery to negative terminal) = negative
- iii) If we go through the resistor in the same direction as current, then there is a fall in potential.
Hence this voltage is taken as negative.
- iv) If we go through the resistor against the direction of current then there is a rise in potential.
Hence this voltage drop is taken as positive.

Example: Write the loop equation for the given circuit shown below

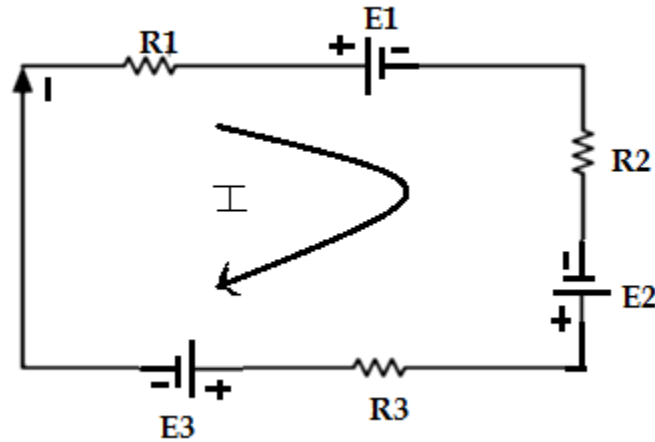


Illustration of Kirchhoff Voltage Law

Applying KVL to the loop we can write

$$-E_1 - I R_2 + E_2 - I R_3 - E_3 - I R_1 = 0$$

$$-E_1 + E_2 - E_3 = I R_1 + I R_2 + I R_3$$

Or

$$E_1 - E_2 + E_3 = -I R_1 - I R_2 - I R_3$$

$$E_1 - E_2 + E_3 = -I (R_1 + R_2 + R_3)$$

1.9 Nodal Analysis

There are two basic methods that are used for solving any electrical network: **Nodal analysis** and **Mesh analysis**. In Nodal analysis, we will consider the node voltages with respect to Ground. Hence, Nodal analysis is also called as **Node-voltage method**.

Procedure of Nodal Analysis

Follow these steps while solving any electrical network or circuit using Nodal analysis.

Step 1: Choose a reference node to which all node voltages can be referred. Label all the other nodes with (unknown) values of voltage, V_1 , V_2 , etc.

Step 2: Assign currents in each connection to each node, except the reference node, in terms of the node voltages, V_1 , V_2 , etc.

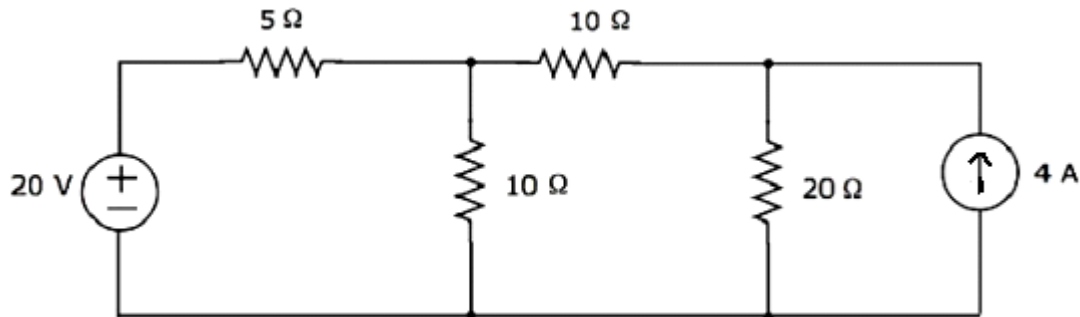
Step 3: Apply Kirchhoff's current law at each node, obtaining as many equations as there are unknown node voltages.

Step 4: Solve the resulting equations to find the node voltages.

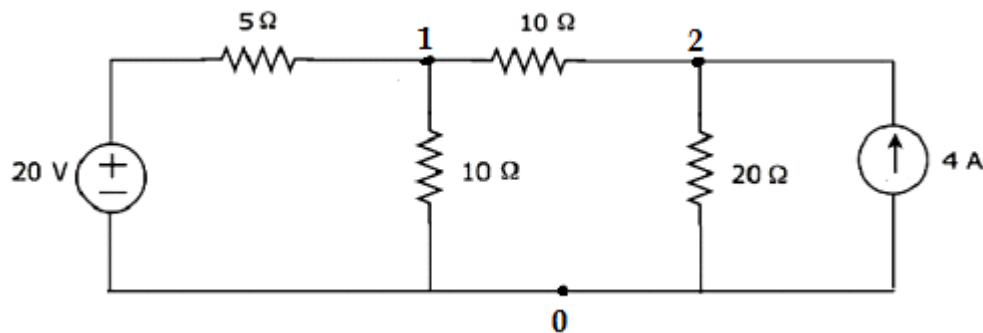
Now, we can find the current flowing through any element and the voltage across any element that is present in the given network by using node voltages.

Example

Find the current flowing through $20\ \Omega$ resistor of the following circuit using Nodal analysis.

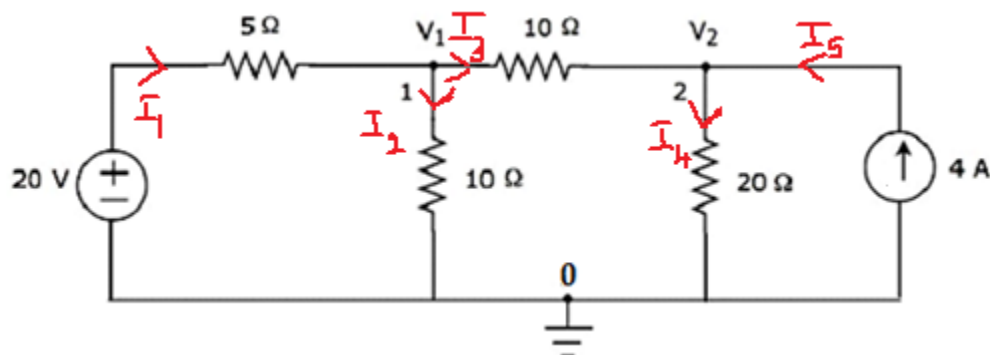


Step 1 – There are three principle nodes in the above circuit. Those are labelled as 1, 2, and 3 in the following figure.



In the above figure, consider **node 0** as reference node (Ground).

Step 2 – The node voltages, V_1 and V_2 , are labelled in the following figure.



In the above figure, V_1 is the voltage from node 1 with respect to ground and V_2 is the voltage from node 2 with respect to ground.

Step 3 – In this case, we will get **two nodal equations**, since there are two principal nodes, 1 and 2, other than Ground.

The **nodal equation** at node 1 is

$$\Rightarrow I_1 - I_2 - I_3 = 0$$

$$\Rightarrow I_1 = I_2 + I_3$$

$$\Rightarrow \frac{20 - V_1}{5} = \frac{V_1}{10} + \frac{V_1 - V_2}{10}$$

$$\Rightarrow 4 - 0.2 V_1 = 0.1 V_1 + 0.1 V_1 - 0.1 V_2$$

$$\Rightarrow 0.4 V_1 - 0.1 V_2 = 4 \text{ ----- (1)}$$

The **nodal equation** at node 2 is

$$I_3 - I_4 + I_5 = 0$$

$$\Rightarrow I_3 + I_5 = I_4$$

$$\Rightarrow \frac{V_1 - V_2}{10} + 4 = \frac{V_2}{20}$$

$$\Rightarrow 0.1 V_1 - 0.1 V_2 + 4 = 0.05 V_2$$

$$\Rightarrow 0.1 V_1 - 0.15 V_2 = -4 \text{ ----- (2)}$$

Step 4 – Find node voltages, V_1 and V_2 by solving Equation 1 and Equation 2.

$$\Rightarrow V_1 = 20V$$

$$\Rightarrow V_2 = 40V$$

So, we got the node voltages V_1 and V_2 as **20 V** and **40 V** respectively.

Step 5 – The voltage across 20Ω resistor is nothing but the node voltage V_2 and it is equal to 40 V. Now, we can find the current flowing through 20Ω resistor by using Ohm's law.

$$I_{20\Omega} = \frac{V_2}{R}$$

Substitute the values of V_2 and R in the above equation.

$$I_{20\Omega} = \frac{40}{20}$$

$$\Rightarrow I_{20\Omega} = 2A$$

Therefore, the current flowing through 20Ω resistor of given circuit is **2 A**.

Note – From the above example, we can conclude that we have to solve 'n' nodal equations, if the electric circuit has 'n' principal nodes (except the reference node). Therefore, we can choose Nodal analysis when the **number of principal nodes** (except reference node) is less than the number of meshes of any electrical circuit.

1.10 Mesh Analysis

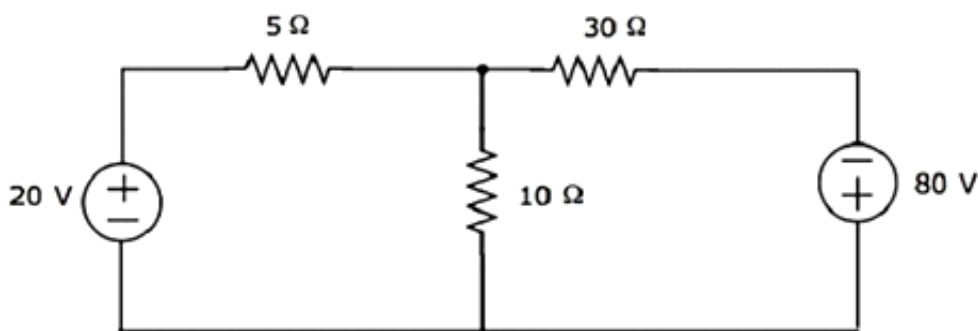
Follow these steps while solving any electrical network or circuit using Mesh analysis.

- **Step 1** – Identify the **meshes** and label the mesh currents in either clockwise or anti-clockwise direction.
- **Step 2** – Observe the amount of current that flows through each element in terms of mesh currents.
- **Step 3** – Write **mesh equations** to all meshes. Mesh equation is obtained by applying KVL first and then Ohm's law.
- **Step 4** – Solve the mesh equations obtained in Step 3 in order to get the **mesh currents**.

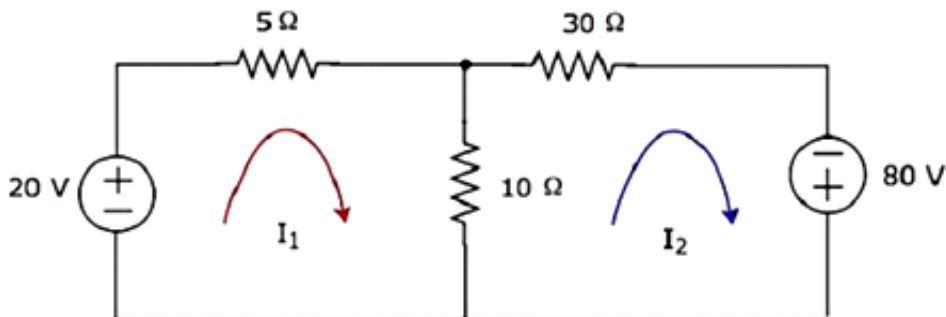
Now, we can find the current flowing through any element and the voltage across any element that is present in the given network by using mesh currents.

Example

Find the voltage across $30\ \Omega$ resistor using **Mesh analysis**.



Step 1 – There are two meshes in the above circuit. The **mesh currents** I_1 and I_2 are considered in clockwise direction. These mesh currents are shown in the following figure.



Step 2 – The mesh current I_1 flows through 20 V voltage source and $5\ \Omega$ resistor. Similarly, the mesh current I_2 flows through $30\ \Omega$ resistor and -80 V voltage source. But, the difference of two mesh currents, I_1 and I_2 , flows through $10\ \Omega$ resistor, since it is the common branch of two meshes.

Step 3 – In this case, we will get **two mesh equations** since there are two meshes in the given circuit. When we write the mesh equations, assume the mesh current of that particular mesh as greater than all other mesh currents of the circuit.

The **mesh equation** of first mesh is

$$\begin{aligned} 20 - 5I_1 - 10(I_1 - I_2) &= 0 \\ \Rightarrow 20 - 15I_1 + 10I_2 &= 0 \\ \Rightarrow 3I_1 - 2I_2 &= 4 \quad \text{-----(1)} \end{aligned}$$

The **mesh equation** of second mesh is

$$\begin{aligned} 80 - 10(I_2 - I_1) - 30I_2 &= 0 \\ \Rightarrow 80 - 40I_2 + 10I_1 &= 0 \\ \Rightarrow I_1 - 4I_2 &= -8 \quad \text{-----(2)} \end{aligned}$$

Step 4 – Finding mesh currents I_1 and I_2 by solving Equation 1 and Equation 2.

$$\Rightarrow I_1 = 3.2 \text{ A}$$

$$\Rightarrow I_2 = 2.8 \text{ A}$$

So, we got the mesh currents I_1 and I_2 as **3.2 A** and **2.8 A** respectively.

Step 5 – The current flowing through 30Ω resistor is nothing but the mesh current I_2 and it is equal to **2.8 A**. Now, we can find the voltage across 30Ω resistor by using Ohm's law.

$$V_{30\Omega} = I_2 R$$

Substitute the values of I_2 and R in the above equation.

$$V_{30\Omega} = 2.8 * 30$$

$$\Rightarrow V_{30\Omega} = \mathbf{84 \text{ V}}$$

Therefore, the voltage across 30Ω resistor of the given circuit is **84 V**.

Note 1 – From the above example, we can conclude that we have to solve 'm' mesh equations, if the electric circuit is having 'm' meshes. That's why we can choose Mesh analysis when the number of meshes is less than the number of principal nodes (except the reference node) of any electrical circuit.

Note 2 – We can choose either Nodal analysis or Mesh analysis, when the number of meshes is equal to the number of principal nodes (except the reference node) in any electric circuit.